Bresenhum's Line Algorithm:

x1,y1=8,5
x2,y2= 12,8

**1st Step: find the slope**: m=y2-y1/x2-x1
m=8-5/12-8=0.75

**2nd Step: Decision parameter, P0= 2dy-dx**

p0= 2*3-4=2

**3rd step:**

| **m<1** | | **m>=1** | |
|---|---|---|---|
| **p<0** | **p>=0** | **p<0** | **p>=0** |
| **Xi+1=Xi+1** | **Xi+1=Xi+1** | **Xi+1=Xi (no change)** | **Xi+1=Xi+1** |
| **Yi+1=Yi (no change)** | **Yi+1=Yi+1** | **Yi+1=Yi+1** | **Yi+1=Yi+1** |
| **P1=p0+2dy** | **P1=p0+2dy-2dx** | **P1=p0+2dx** | **P1=p0+2dx-2dy** |

p1=2+2*3-2*4=0
p2=0+6-8=-2
p3=-2+6=4

| p | xi | yi | xi+1 | yi+1 |
|---|---|---|---|---|
| 2 | 8 | 5 | 9 | 6 |
| 0 | 9 | 6 | 10 | 7 |
| -2 | 10 | 7 | 11 | 7 |
| 4 | 11 | 7 | 12 | 8 |
| | 12 | 8 | | |